A Complex Numbers

Definition, and useful properties

A complex number $\tilde{z}$ may be expressed in the form

$$a + ib \tag{A.1}$$

where a and b are real numbers, and $i^2 = -1$. a is called the real part of $\tilde{z}$, and b is its imaginary part.

Complex numbers may be represented by points, or vectors, in a plane, with a real axis and an imaginary axis (Figure A.1).

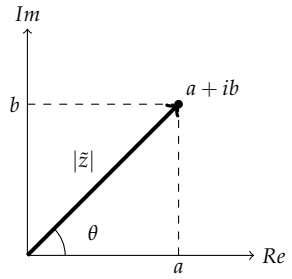


Figure A.1: Representing a complex number in the complex plane.

Complex numbers can be added and multiplied using the same rules that we use for real numbers. For example,

$$(a + bi)(c + di) = ac - bd + i(ad + bc) \tag{A.2}$$

The complex conjugate of $\tilde{z}$, conventionally represented by $\tilde{z}^*$, is constructed by changing i to $-i$ wherever it appears in $\tilde{z}$. So if $\tilde{z} = a + bi$, $\tilde{z}^* = a - bi$.

The magnitude of a complex number, $|\tilde{z}|$, is calculated as $\sqrt{\tilde{z}\tilde{z}^*}$. If we write $\tilde{z}$ as $a + bi$, this means that

$$|\tilde{z}| = \sqrt{(a + bi)(a - bi)} = \sqrt{a^2 + b^2} \tag{A.3}$$

which is what we would expect for a vector in a plane.

Just like vectors in a plane, we can also represent a complex number using polar coordinates:

$$\tilde{z} = |\tilde{z}|(\cos\theta + i\sin\theta) \tag{A.4}$$

An equivalent expression makes use of the amazing Euler identity:

$$e^{i\theta} = \cos\theta + i\sin\theta \tag{A.5}$$

which allows us to write

$$\tilde{z} = |\tilde{z}|e^{i\theta} \tag{A.6}$$

In particular, note that the imaginary number i can be written as $e^{i\pi/2}$.

The conversion relations between Cartesian and polar coordinates are

$$a = |\tilde{z}| \cos\theta \tag{A.7a}$$
$$b = |\tilde{z}| \sin\theta \tag{A.7b}$$

and

$$|\tilde{z}| = \sqrt{a^2 + b^2} \tag{A.8a}$$
$$\theta = \arctan\left(\frac{b}{a}\right) \tag{A.8b}$$

A caveat about applying equation (A.8b): There are two possible values of θ that satisfy the equation, and they differ by π. So you must look at the signs of the real and imaginary parts of $\tilde{z}$ to see which quadrant $\tilde{z}$ is in, to decide which value of θ is the right one.

If a complex number $\tilde{z} = |\tilde{z}|e^{i\theta}$ is multiplied by $e^{i\phi}$, the operation has the effect of rotating $\tilde{z}$ through an angle ϕ. This is quite simply because

$$(|\tilde{z}|e^{i\theta})e^{i\phi} = |\tilde{z}|e^{i(\theta+\phi)} \tag{A.9}$$

In particular, to multiply by i is equivalent to rotating the vector by $\pi/2$. If we multiply by i again, the vector is rotated by π, which turns out to be the negative of the original vector. This is what we mean by the statement $i^2 = -1$!

Example 1.1: *$\tilde{z}_1 = 3.0e^{i\pi/8}$ and $\tilde{z}_2 = 4.0e^{-i\pi/3}$. What is the sum $\tilde{z} = \tilde{z}_1 + \tilde{z}_2$ in cartesian and in polar coordinates?*

First convert the two complex numbers to a cartesian representation:

$$\tilde{z}_1 = 3.0(\cos\pi/8 + i\sin\pi/8) = 2.77 + 1.15i$$
$$\tilde{z}_2 = 4.0(\cos(-\pi/3) + i\sin(-\pi/3)) = 2.00 - 3.46i$$

Next, add the two numbers to get z in cartesian form:

$$\tilde{z} = 4.77 - 2.32i$$

The phase angle of $\tilde{z}$ is

$$\phi = \arctan\left(\frac{4.77}{-2.32}\right) = -0.452\,\text{rad} = -25.9°$$

This corresponds to a point in the 4th. quadrant, which is consistent with the real and imaginary parts of $\tilde{z}$ calculated before.

Finally, the magnitude of $\tilde{z}$ is

$$|\tilde{z}| = \sqrt{(4.77)^2 + (-2.32)^2} = 5.30$$
